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Environmental impacts and cost of a water quality trading approach for NPDES nutrient permit compliance in a rural watershed

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ABSTRACT

U.S. EPA's National Water Quality Trading (WQT) policy encourages market-based approaches to improve water quality and maximize ancillary benefits beyond the primary purpose of water quality permit compliance. Wisconsin has a growing WQT program through which nutrient point sources such as wastewater treatment facilities can purchase nutrient reduction credits generated by other entities within the watershed, such as agricultural producers who convert cropland to grassland or forest. However, benefits beyond nutrient reduction are seldom quantified, suggesting the full benefits may be underestimated. This project uses life cycle assessment and life cycle cost assessment to characterize the economic and environmental impacts of a WQT approach to nutrient compliance, using a rural watershed in the Wisconsin driftless area as a case study. The results demonstrate WQT is generally a lower cost means of achieving NPDES permit compliance. The results also suggest additional benefits with WQT, including reductions in nutrient emissions beyond permit requirement, increased soil carbon, and reduced energy demand. The results not only demonstrate the benefits of integrated watershed management, but also show how, compared to a technological approach, a nature-based solution can achieve water quality compliance at a lower overall cost. It sheds light on incentivizing potential broader adoption of WQT.

1. Introduction

Under the U.S. Clean Water Act, National Pollutant Discharge Elimination System (NPDES) permits and Total Maximum Daily Loads (TMDL) programs are developed by states and tribes to reduce pollutant discharges and meet numeric water quality criteria. In addition to traditional compliance approaches, the NPDES program encourages water quality trading (WQT) to allow for flexibility in meeting water quality improvement goals (U.S.EPA 2003; U.S.EPA 2019). As an alternative to installing technological infrastructure upgrades, WQT allows dischargers to purchase pollution reduction credits to comply with regulatory requirements such as low nutrient effluent standards. Credit generators are those entities that can achieve, through alternative means, the required level of pollution reduction at a lower cost than the credit purchaser to achieve equivalent environmental outcomes.

Even though the U.S. Environmental Protection Agency (EPA) issued its WQT Policy in 2003, WQT in U.S. has not been widely adopted. Some of the reasons for the limited adoption of WQT include high transaction costs, thin markets (inadequate demand and supply of credits), uncertainty in measuring non-point source reductions, not enough feasible watersheds for trading to occur, inadequate incentives to participate in the market, among many challenges (Hoag et al., 2017; Heberling et al., 2018). In 2020, U.S. EPA provided further guidance in *Water Quality Trading on a Watershed Scale* (U.S.EPA 2020), in which it describes factors to consider when evaluating the feasibility of WQT. These include ensuring the program implemented at a scale that is informed by the hydrology and the ecology of the watershed and that sufficient supply and demand of credits exist. Wisconsin has a growing WQT program that starts to exhibit these characteristics. As part of the program, wastewater treatment facilities (WWTFs) can purchase nutrient credits in lieu of

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making upgrades to their treatment processes (WDNR 2024). Credits are typically generated within the same watershed through practices that reduce nonpoint source nutrient pollution. In 2024, the state implemented a WQT Clearinghouse to further facilitate and standardize trading between buyers and sellers (Wisconsin Clearinghouse, 2024).

Alongside growing interest in WQT, many states are adopting more stringent NPDES nutrient permit limits in response to increasing waterbody eutrophication. Wisconsin, in particular, has one of the most stringent statewide effluent standards in the US (Meyer and Raff, 2024). For small or rural community WWTFs, these standards may be particularly difficult for their WWTFs to meet due to limited revenue bases. WQT may offer a lower cost alternative.

Critics of WQT programs note that WQT only results in a reduction in compliance costs by design, not in further reductions to nutrient emissions beyond what is required by local standards (Stephenson and Shabman, 2017). However, in the case of small, rural communities that cannot afford to immediately upgrade their WWTF, simply meeting the standard is, arguably, preferable to years of inaction and non-compliance. Additionally, most non-point source projects that generate nutrient credits may confer other benefits to the watershed that, although not all currently monetizable, can be environmentally beneficial.

The purpose of this study is to evaluate the economic and environmental benefits of a WQT approach to nutrient compliance in a case study watershed in Wisconsin using nine small WWTFs in Wisconsin's rural, predominantly agricultural Kickapoo River watershed. These WWTFs have effluent phosphorus concentrations of between 1 and 5 mg/L, compared to in-stream standards 0.075 to 0.10 mg/L (Wisconsin State, 2022). For each WWTF, three scenarios are modeled, including a baseline (existing condition), a standard technological upgrade, and a WQT scenario. The WQT approach is the long-term conversion of corn and soybean croplands to perennial grassland. Life cycle assessment (LCA) and life cycle cost assessment (LCCA) are used to compare these scenarios over a 30-year time horizon. LCA results are based, in part, on the results of Wisconsin's Soil Nutrient Application Planner (SnapPlus) and Soil & Water Assessment Tool (SWAT) models, both of which are based on current or recent historical hydrologic conditions of the study watershed. To streamline the development of life cycle models, representative examples of each WWTF type - activated sludge (AS), lagoon and rotating biological contactor (RBC) are used to develop detailed system archetypes that can be used to represent the characteristics of the other six WWTFs. These archetype models are based on characteristics specific to the actual facilities, taken largely from publicly available engineering reports. This work demonstrates how holistic analysis with integrated metrics can shed lights on whether potential new policy options using market-based approaches can achieve better environmental outcomes at lower cost.

2. Results and discussion

2.1. WQT area results

To compare WQT to a conventional technological upgrade, the reduction in mass loading of effluent phosphorus required by each WWTF (i.e., the credit need) was first calculated (Table 1). Table A1 lists the range of options considered for the three archetype WWTFs, which are also representative of the options considered at the other WWTFs (details in 4.4.1, 4.6 and Appendix A). SnapPlus, a software program that Wisconsin credit generators must use to calculate credits, was then used to calculate the comparable amount of cropland conversion needed to generate sufficient credits. The credit generating potential of existing cropland varies as a function of slope, soil phosphorus content, soil texture and management activities such as fertilizer application, grazing, tillage practices and vegetation type.

Table 1
Water Quality Trading Phosphorus Credit Needs and Land Area Requirements.

Permittee	WWTF Credit Need[1] kg/yr (lb/yr)	Areal Average TP Reduction[2] kg/ha/yr (lb/ac/yr)	Required Field Area for Trading[3] ha (acre)
Ontario WWTF	259 (572)	2.40 (2.14)	130 (320)
Viola WWTF	277 (610)	1.56 (1.44)	213 (526)
Wauzeka WWTF	182 (402)	2.9 (2.59)	75 (186)
Gays Mills WWTF	179 (395)	2.36 (2.11)	91 (225)
La Farge WWTF	97 (214)	4.92 (4.39)	24 (59)
Norwalk WWTF	193 (426)	1.50 (1.34)	155 (382)
Readstown WWTF	115 (254)	5.04 (4.50)	27 (68)
Soldiers Grove WWTF	228 (502)	3.30 (2.94)	83 (205)
Wilton WWTF	326 (719)	1.14 (1.02)	342 (844)

1) Calculated using Eq. (1). 2) SnapPlus results for average reduction in TP export from selected WQT fields realized by converting from existing crop to perennial grass. 3) Calculated as 1.2*Credit Need/TP Reduction (Minimum trading ratio of 1.2 is required to account for uncertainty. Details in 4.3.2).

2.2. Nutrient export results

Because SnapPlus only focuses on phosphorus and we are interested in total nutrient emissions, we used SWAT to model changes in all nutrient export including total nitrogen (TN) from converted croplands (Table 2). More information about the SWAT modeling approach and model inputs are provided in Appendix C and Section 4.3. Using both models also allows for a comparison of phosphorus export results.

Although considerable variability exists across both fields and models, both models estimate TP reductions of $>$ 90 % can be achieved from conversion of intense tillage fields to perennial grassland. SnapPlus generally provides higher overall export estimates than SWAT, which translates to higher estimates of credit generation upon cropland conversion. For intense tillage scenarios, SnapPlus predicted watershed-average exports of 1.3–5.2 (3.0 average) kg/ha/yr, while SWAT predicted 1.3–2.9 (2.2 average). Combined with average exports from grassland of 0.19 and 0.10 kg/ha/yr, resulting average reductions are 2.8 and 2.1 kg/ha/yr from SnapPlus and SWAT, respectively.

Both SnapPlus and SWAT simulate long-term nutrient dynamics, but the two are inherent different. To predict soil erosion dynamics, SnapPlus utilizes the Revised Universal Soil Loss Equation, Version 2 (RUSLE2), whereas SWAT utilizes the Modified Universal Soil Loss Equation (MUSLE). Both are versions of the USLE, however the RUSLE2 model measures soil loss per unit area at an annual timescale using the same rainfall characteristics each year, while MUSLE uses daily rainfall runoff and peak flow rate to estimate event-based soil loss at a finer temporal resolution (Sadeghi et al. 2014; Benavidez et al. 2018). Given the nonlinear relationship between rainfall intensity and soil erosion, SWAT results have considerable interannual variability (Table 2, Fig. 1). Interannual variability in SnapPlus is only driven by cropping type and annual TP losses, not climate variability. Also, SWAT-estimated P loss is more influenced by mineral and organic P loss associated with erosion than dissolved P loss in runoff. Therefore, variability across fields (Table 2) is driven by rainfall, soil properties and slope variations.

From field research conducted in Wisconsin and Minnesota, which examined corn-soybean crop rotations, a wide range of tillage methods, and alternative fertilizer sources (manure and synthetic P fertilizer) has shown that measured annual average total phosphorus losses at the field's edge can vary from 0.1 kg/ha to 3.5 kg/ha (Good et al. 2012). Total phosphorus (TP) loss from corn silage farms in the Jersey Valley watershed of Wisconsin, a hydrologically similar area to the Kickapoo watershed, was measured to range from 0.97 to 1.92 kg/ha/year (Radatz et al. 2018).

Table 2
Average Annual Nutrient Export Results Across WQT Fields.

Permittee	SNAPPLUS - TP			SWAT - TP			SWAT - TN		
	Int. Tillage[1] kg/ha/yr (lb/ac/yr)	Per. Grass[2]	Red.[3]	Int. Tillage	Per. Grass	Red.	Int. Tillage	Per. Grass	Red.
Ontario	2.6 (2.3)	0.2 (0.1)	2.4 (2.1)	1.8 (1.6)	0.1 (0.1)	1.6 (1.5)	14 (12)	10 (9)	3.8 (3.4)
Viola	1.9 (1.7)	0.3 (0.3)	1.6 (1.4)	2.5 (2.2)	0.1 (0.1)	2.3 (2.1)	16 (14)	11 (10)	4.8 (4.3)
Wauzeka	3.0 (2.7)	0.1 (0.1)	2.9 (2.6)	1.7 (1.6)	0.0 (0.0)	1.7 (1.5)	14 (12)	10 (9)	3.4 (3)
Gays Mills	2.5 (2.2)	0.1 (0.1)	2.4 (2.1)	2.9 (2.6)	0.1 (0.1)	2.8 (2.5)	17 (15)	11 (10)	5.2 (4.6)
La Farge	5.1 (4.6)	0.2 (0.2)	4.9 (4.4)	2.6 (2.3)	0.2 (0.2)	2.4 (2.1)	17 (15)	11 (10)	5.7 (5.1)
Norwalk	1.7 (1.5)	0.2 (0.2)	1.5 (1.3)	2.3 (2.1)	0.1 (0.1)	2.2 (2.0)	16 (14)	10 (9)	5.6 (5)
Readstown	5.2 (4.6)	0.2 (0.1)	5.0 (4.5)	2.5 (2.3)	0.1 (0.1)	2.4 (2.2)	16 (15)	12 (10)	4.7 (4.2)
Soldiers Grove	3.5 (3.1)	0.2 (0.2)	3.3 (2.9)	1.3 (1.1)	0.0 (0.0)	1.2 (1.1)	18 (16)	12 (11)	5.5 (4.9)
Wilton	1.3 (1.2)	0.2 (0.2)	1.1 (1.0)	2.4 (2.2)	0.1 (0.1)	2.3 (2.1)	14 (12)	10 (9)	3.6 (3.2)
Average:	**3.0 (2.7)**	**0.2 (0.2)**	**2.8 (2.5)**	**2.2 (2)**	**0.1 (0.1)**	**2.1 (1.9)**	**16 (14)**	**11 (10)**	**4.7 (4.2)**

1-Intense tillage is considered "Fall Chisel, Disked" in SnapPlus, which includes a fall chisel plowing with spring disking and field cultivation before planting. 2-Perrenial grassland does not include tillage or fertilizer. 3-Reduction is the difference between intense tillage and perennial grassland.

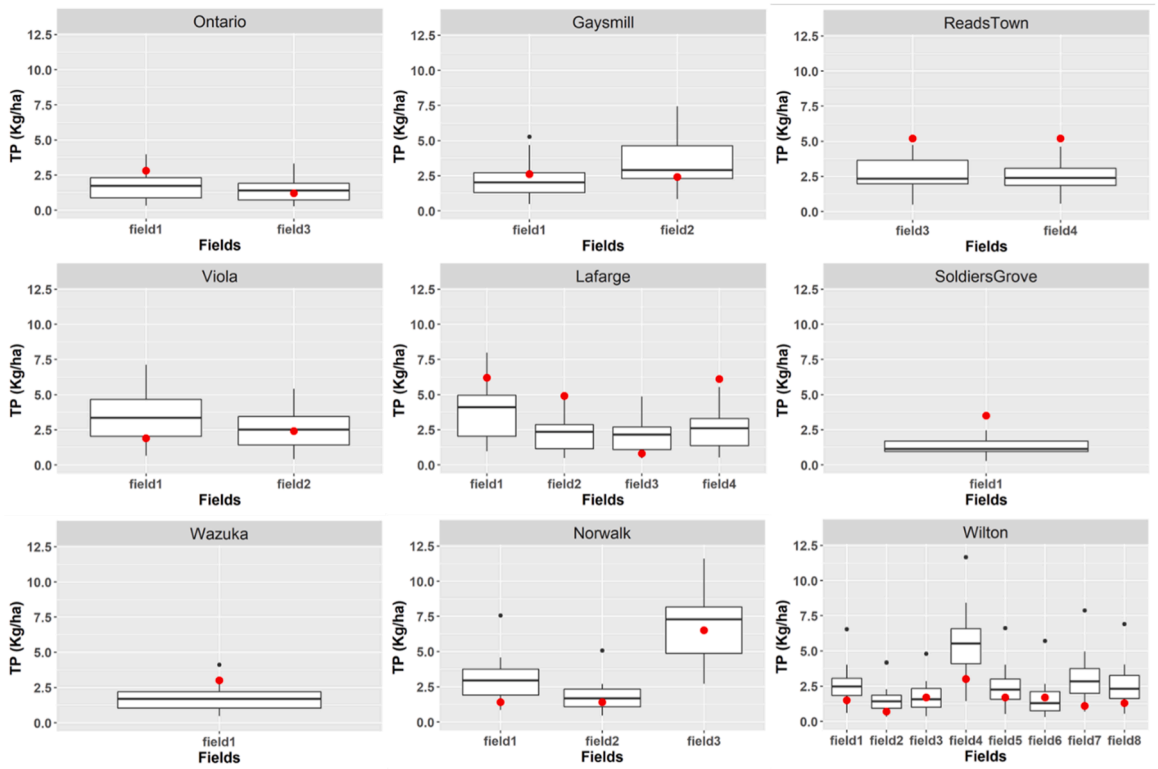


Fig. 1. Annual variation in TP load estimated by SWAT (box and whisker) compared to annual average TP load estimated by SnapPlus (red dot) for all modeled fields.

2.3. Life cycle assessment results

For each WWTF, life cycle inventories were developed for the baseline, technological upgrade and WQT scenarios over a 30-year time horizon. Detailed life cycle inventories (LCIs) were developed for the three archetype facilities and extrapolated to the other facilities based on system size and type. The technological upgrade scenario consisted of adding sand filtration and chemical coagulation to existing treatment processes, while the baseline and WQT scenarios included and excluded, respectively, use of diesel fuel and synthetic fertilizers for typical farming activities. LCI development and life cycle impact assessment (LCIA) modeling is discussed in Sections 4.4 and 4.5, and detailed model inputs are provided in Appendix A.

2.3.1. Eutrophication potential

Eutrophication potential (EP) assesses the cumulative potential impact from eutrophying compounds, including nitrogen, phosphorus and BOD, on receiving environments (Bare et al. 2002; Bare 2011). EP results for this study, are driven by effluent and crop production emissions for the baseline scenarios, which decrease in the treatment upgrade and WQT scenarios (Fig. 2a). In most cases, WQT scenarios perform similar to or better than the technological upgrade scenarios mostly due to incorporation of a trade ratio of 1.2 (all else equal, the trade ratio ensures more reduction will be achieved upstream than is needed at the WWTF) (details in 4.3.2) and the additional reduction in nitrogen loading from cropland conversion (the combination of chemical coagulation and rapid sand filtration employed in the technological upgrade scenario only reduces phosphorus in WWTF effluent). The LCA eutrophication potential includes both nitrogen and phosphorus emission potentials.

Comparing all archetype WWTF performance (Fig. 2b), the variability in nutrient modeling (Table 2) can lead to conditions where the

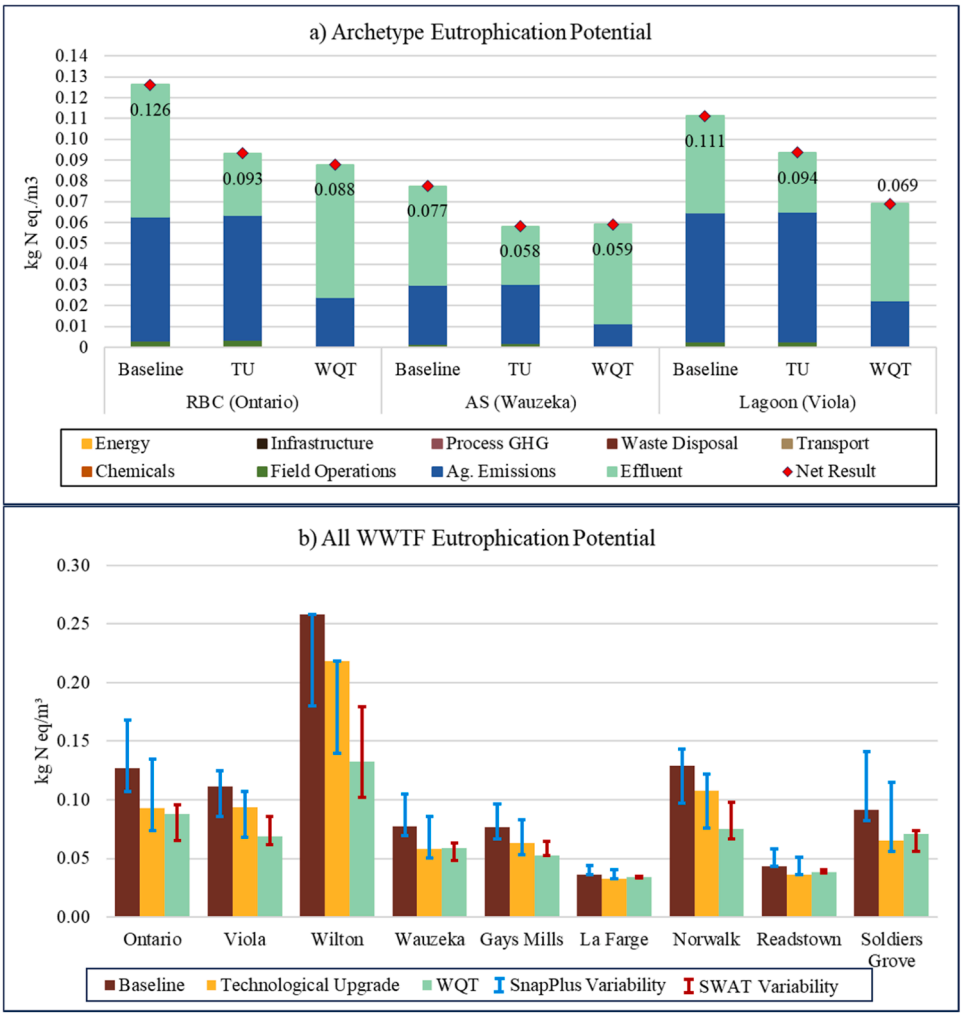


Fig. 2. Eutrophication potential results for archetype systems (a) and all WWTFs (b). AS – activated sludge. RBC – rotating biological contactor. TU – technological upgrade. WQT – water quality trading.

WQT and technology upgrade scenarios are more comparable. Notable examples include La Farge and Readstown. SnapPlus results for these two WWTFs suggest TP reductions from cropland conversion would be the highest among all WWTFs at 4.9 and 5.0 kg/ha/yr (Table 2), which translates to the fewest acres required to satisfy their credit needs among all WWTFs (Table 1). However, estimated TP reductions from SWAT are roughly half those predicted by SnapPlus for these fields. For the corresponding WWTFs, fewer trading acres combined with potentially lower actual nutrient reductions from those trading fields lead to lower EP reductions. Although the EP of the WQT scenarios for these two WWTFs is always less than that of the baseline scenarios, even including SnapPlus and SWAT variability ranges (Fig. 2b), these results illustrate SnapPlus could overestimate potential TP reductions for WQT fields.

Conversely, Wilton illustrates what happens when SnapPlus TP reduction estimates are low, corresponding high trading acre, and SWAT nutrient reductions are higher than those predicted by SnapPlus (Table 2) – EP reductions modeled for the WQT scenario of Wilton WWTF are the highest of the group.

2.3.2. *Climate change potential*

Climate change potential (CCP) quantifies the potential heat trapping capacity of greenhouse gases emitted by the construction and operation of study systems over their life cycle (Bare et al. 2002; Bare 2011). The RBC and lagoon-based archetypes have similar CCP in both the baseline and technological upgrade scenarios (Figure A4a). The lagoon system has a lower energy demand than the RBC process but accrues a considerable CCP impact associated with use of sulfuric acid in the neutralization process. The activated sludge archetype has the highest CCP of all three archetypes across all three scenarios due to its high energy demand for aeration.

The additional energy requirements of chemical coagulation and rapid sand filtration lead to higher CCP impacts in the technological upgrade scenarios relative to both the baseline and WQT scenarios (Figure A4a). WQT scenarios result in the lowest CCP for all WWTFs due to the cessation of agricultural field operations and the increased potential for soil carbon sequestration upon conversion to grassland. The higher average credit per unit area generated by the fields in Wauzeka (Table 2) leads to a smaller land area requirement to meet credit requirements and a correspondingly smaller CCP reduction from cessation of agricultural activities and smaller carbon sequestration associated the land conversion.

Across all other WWTFs, WQT scenarios always result in a net benefit to CCP, even across the variability possible in SnapPlus results (Figure A4b). Similar to EP, the benefits are greatest for Wilton, where SnapPlus provided low estimates of TP reduction from cropland conversion, resulting in high trading acre requirements. With more acres converted from crop to grassland, more CCP benefits are realized by Wilton.

2.3.3. Cumulative energy demand

Cumulative energy demand (CED) quantifies the total energy inputs, including renewable and non-renewable, to the study systems over their life cycle. CED is highest for the activated sludge archetype across all scenarios (Figure A5a). The additional energy requirements of chemical coagulation and rapid sand filtration lead to higher CED in the technological upgrade scenarios relative to both the baseline and WQT scenarios. Cessation of field operations is all that affects energy demand in the land conversion, leading to equivalent energy demand between the baseline WWTF and WQT scenarios. For this same reason, there are no variability ranges attributable to the WQT scenario for each WWTF (Figure A5b). Still, across all potential variability in SnapPlus results, CED for WQT scenarios is lowest for all modeled WWTFs (Figure A5b).

2.4. Life cycle cost assessment results

LCCA results for the three treatment archetypes show the cost per m^3 of treated effluent, as net present value (NPV), necessary to meet phosphorus effluent requirements using either technological upgrades or WQT (Fig. 3a). WQT has the lowest cost for all three archetypes. The largest savings are realized by the activated sludge archetype (Wauzeka). The factors that contribute to this relatively better performance include less credit needs than the RBC (Viola) archetype (see Tables 1 and 3) and having WQT fields with greater TP reduction potential than the other two archetype systems (Table 1).

Another contributing factor to the difference is the ratio of average annual flow rate to design flow rate specific to each community. The lagoon archetype (Viola) operates closest to its design flow rate (Table 3) and therefore spreads out the cost of the technological upgrade over a larger quantity of treated water, resulting lower cost per volume than RBC and AS upgrades. Additionally, land area requirements in the WQT scenario are designed to match the annual average flow rate, contributing to its lower cost per m^3 treated. It is likely easier to match WQT credit purchases to average flow rates than construct overcapacity technological upgrades needed for future growth. Also, most costs for the technological upgrade scenario are capital costs whereas annual credit payments constitute the bulk of life cycle costs for the WQT scenario. This difference affects the timing of funding requirements and should be considered when weighing similar compliance alternatives.

When expanding archetype costs to other WWTFs (Fig. 3b), results show that Wilton is the only facility where the WQT scenario is more expensive than the technological upgrade scenario. Of all WWTFs, Wilton has the largest gap between existing and target effluent TP concentrations (Table 3) and therefore requires more WQT acres to meet their credit need. Conversely, chemical coagulation for TP removal is capital intensive but once facilities are in place, additional TP removal can be achieved through proportionally higher doses of relatively inexpensive coagulant. Wilton is also a lagoon facility, based on the archetype analysis, is the most inexpensive system type across technological upgrade scenarios. If Wilton were an RBC or AS facility, the cost of WQT would likely be more comparable to that of the technological upgrade scenario, especially considering the variability associated with SnapPlus results.

2.5. Discussion

Critics of WQT programs argue these programs cannot truly address the nonpoint source nutrient problem, as market-based approaches simply trade control of one source for another with no net change in nutrient emissions and pollutant load (Stephenson and Shabman 2017). This study's results do not support this claim. With just phosphorus as the regulated pollutant, the trade ratio is the only systematic mechanism by which slightly more phosphorus may be removed in a WQT approach than a technological upgrade. However, when the total eutrophication from both nitrogen and phosphorus are considered, the combination of the trade ratio and nitrogen reduction result in, on average, lower eutrophication potential to the WWTF watersheds under a WQT approach (Fig. 2). Anecdotally, the Wisconsin Department of Natural Resources (WDNR) has also found that allowing point sources to consider WQT as a compliance option (e.g., WDNR 2020a; WDNR 2023) has accelerated implementation of their waste load allocation (Azevedo 2024). Additionally, depending on the practice that generates credits (Wisconsin allows for other practices such as riparian buffers and streambank erosion mitigation), nutrient reduction may be just one of many environmental benefits. We therefore think of WQT not as a 'silver bullet' solution but as a means of facilitating lower-cost nutrient compliance for small WWTFs that can simultaneously incentivize farmers to transition towards practices with broader environmental benefits.

In 2020, the Wisconsin Governor's Task Force on Climate Change Report recommended to develop a program to pay farmers to increase soil carbon storage in agricultural lands through tax incentives or subsidies (State of Wisconsin 2020). While, to our knowledge, no such program exists at the state level, there are private companies that currently facilitate carbon trading between farmers and businesses to reduce their carbon footprint. Indigo Ag is one example and pays approximately $0.08/kg of carbon sequestered for each year the farm participates in the program (Indigo 2023). Using our assumed rate of carbon sequestration due to cropland conversion of 76 kg/ha/yr and extending to a 30-year NPV, potential revenue for each facility or landowner would average $0.16/$m^3$ ($0.02 to $0.54/$m^3$) if normalized to the volume of wastewater treated. These revenues are of the same order of magnitude of the WQT costs discussed in Section 3.4 ($0.41 to $0.73/$m^3$), representing a potentially meaningful additional return to farmers already participating in a WQT market.

This study design does not address the loss of crop production that would occur upon conversion of cropland to grassland. On a small scale, this may be a reasonable assumption as the yield contribution of hundreds of acres is unlikely to affect supply/demand balances of corn and soybean markets. However, the larger the scale, the more likely it is this type of program will have a non-trivial influence on market dynamics. From a global environmental perspective, if agricultural production is simply shifted to places with worse nutrient and soil management practices, the net impact of such a WQT program could be negative.

One approach to minimize burden shifting is targeting marginal cropland first. Keller and Fox (2019) evaluated the availability of marginal cropland (those with low infiltration rates and high slopes) for reforestation to determine the magnitude of impact a water quality trading program could have in the Ohio River Basin. Of the 17.7 million acres of cropland (32 % of the Ohio River Basin), they found 14 % (4.5 % of the Ohio River Basin) to be in marginally productive areas. If reforested, they estimated it could be 12 % and 5 %, respectively, of annual export of nitrogen and phosphorus from the entire Ohio River Basin .

To establish a successful WQT program in any country or government, the pollution sources and goals need to be clearly established, especially with strong regulatory and non-regulatory drivers. Robust stakeholder engagement is also critical. Ribaudo and Savage (2014) highlighted the risk of WQT benefits being overstated if pre-program conditions are misrepresented, or those changes were likely to occur without a WQT program. Wisconsin's framework, through use of a standardized modeling approach for pre- and post-conversion conditions (SnapPlus) and a third-party clearinghouse show how these risks can be reduced. The market size, in term of the credit suppliers and the credit buyers, is also important. Here again, Wisconsin WQT helps to provide a transparent online registry to minimize transaction cost, and increase the number of transactions (Wisconsin Clearinghouse, 2024).

With more stringent environmental regulatory standards worldwide such as the recast Urban Wastewater Treatment Directive in Europe (Office Journal of the European Union, 2024), adaptive approaches like WQT may be needed to achieve ambitious environmental goals.



Fig. 3. NPV, per m^3 of effluent treated, to meet phosphorus effluent requirement via technological upgrades (TU) or water quality trading (WQT) for archetype systems (a) and all WWTFs (b). RBC=rotating.

Table 3
Characteristics of Wastewater Treatment Facilities in the Kickapoo River Basin.

WWTF	Population Served	Biological Process	2016–2020 Average[10]			P WQBEL
			Average Flow (Design Flow)		Effluent P Concentration	
			m^3/d	MGD	mg/L	mg/L
ONTARIO[1]*	554	RBC	156 (326)	0.041 (0.086)	4.6	0.075
VIOLA[2]*	699	Aerated Lagoon	317 (379)	0.084 (0.10)	2.5	0.1
WILTON[3]	504	Aerated Lagoon	162 (337)	0.043 (0.089)	5.6	0.075
WAUZEKA[4]*	250	Activated Sludge	192 (303)	0.051 (0.080)	2.7	0.1
GAYS MILLS[5]	504	Activated Sludge	276 (329)	0.073 (0.087)	1.9	0.1
LA FARGE[6]	746	Activated Sludge	564 (651)	0.15 (0.17)	0.57	0.1
NORWALK[7]	638	Activated Sludge	183 (522)	0.048 (0.14)	3.0	0.075
READSTOWN[8]	415	Activated Sludge	343 (356)	0.091 (0.094)	1.0	0.1
SOLDIERS GROVE[9]	592	Activated Sludge	175 (432)	0.046 (0.11)	3.7	0.1

*Archetypes.
MGD – million gallons per day, P – phosphorus, WQBEL – water quality-based effluent limit, RBC - rotating biological contactor, WWTF – wastewater treatment facility.
[1] (Davy Engineering Co. 2016b).
[2] (Delta 3 Engineering 2016).
[3] (Delta 3 Engineering 2017; Delta 3 Engineering 2019).
[4] (Delta 3 Engineering 2018b).
[5] (Davy Engineering Co. 2019).
[6] (MSA 2017).
[7] (Davy Engineering Co. 2016a).
[8] (Delta 3 Engineering 2018a).
[9] (Davy Engineering Co. 2018).
[10] (Hartenbower 2021).

3. Conclusions

This study's results support the idea that WQT is a cost-effective permit compliance option for small WWTFs. It provides additional environmental benefits including increased soil carbon and reduced eutrophication potential. However, this research is still not

comprehensive. Similar nature-based nutrient management solutions such as reforestation, wetland restoration, and streambank restoration may confer additional benefits, including habitat restoration and flood mitigation, and should be the topic of further research in the context of WQT.

4. Methods

This study presents the results of an LCA and LCCA analysis for nine small WWTFs in the Kickapoo River watershed (Kickapoo watershed). SnapPlus and SWAT were used to define the required water quality trading credits and the change in nutrient loss and erosion that result from converting row crops to perennial grassland, respectively. The analysis complies with the guidelines established for conducting an LCA study in the International Organization for Standardization's (ISO) 14040 and 14044 standards (International Organization for Standardization 2006a; International Organization for Standardization 2006b). The LCCA results were generated using net present value (NPV) methods developed by the National Institute of Standards and Technology (Kneifel and Webb 2022). The detailed descriptions of LCA and LCCA, SnapPlus and SWAT models can be found in Supplemental Information (SI), divided into three Appendices as follows:

- Appendix A – General information, LCA and LCCA
- Appendix B – SnapPlus
- Appendix C – SWAT

4.1. Goal and scope definition

The goal of this study is to compare the environmental and economic impacts of technological and WQT options for small WWTFs to meet increasingly stringent nutrient effluent limitations. The analysis employs a streamlined LCI for baseline and upgraded WWTF technology scenarios and WQT scenarios. The system boundary of the study includes all the primary material and energy inputs, their background supply chain emissions, and direct emissions to the environment associated with operation of the baseline and upgraded WWTFs. The study's functional unit is the treatment of one cubic meter (m^3) of municipal wastewater with phosphorus effluent quality, as defined in Table 3. Results for the WQT scenarios are also presented per m^3 of treated wastewater by standardizing the benefits of grassland conversion to the quantity of wastewater treated annually. Wastewater effluent is discharged to the Kickapoo River or one of its tributaries. Differences in environmental impacts associated with differing effluent water quality are reflected in LCA results and are a key component of scenario definition.

4.2. Scenario descriptions

4.2.1. WWTF baseline scenario

The baseline scenario reflects the current performance of the nine municipal WWTFs in the Kickapoo watershed (Table 3). These are all small facilities serving populations of between 250 and 750 individuals with average flow rates that vary between 0.04 and 0.15 million gallons per day (MGD). Like most facilities in the region, they can be classified according to their main biological process as either activated sludge, lagoon, or rotating biological contactor (RBC) systems. For this study, representative examples of each type were selected and used to develop system archetypes to reflect the treatment performance, cost, and environmental impact of these typical WWTFs in the Kickapoo watershed (they are also likely representative of systems in many small communities across the U.S.).

4.2.2. WWTF upgrade scenario

The upgrade scenario reflects the technological approach necessary to achieve the phosphorus water quality-based effluent limit (WQBEL) in each facility's permit. In response to updated effluent requirements, each WWTF evaluated the feasibility of several technological, natural, and performance-based options to achieve water quality goals. Table A1 lists the range of options considered for the three archetype WWTFs, which are also representative of the options considered at the other WWTFs. For the upgrade scenario, we assume facilities pursued sand filtration with chemical coagulation as it was considered by all facilities and is technologically capable of meeting the WQBELs. While most communities still consider these technologies to be too expensive, it is one of the lower cost technological solutions of all options considered (Table A1).

4.2.3. Water quality trading scenario

The WQT scenario is an alternative to the technological upgrade approach that facilities can use to meet their WQBELs. The Wisconsin Department of Natural Resources (WDNR) allows WWTFs to purchase nutrient reduction credits generated through implementation of nonpoint source management practices, which may include agricultural field and farm land management, streambank stabilization, dredging, wetland restoration, and urban stormwater management (WDNR 2022b). Feasibility analyses included in each WWTF management plan (Table A1) considered WQT. Each case-study community found WQT to potentially be an economically viable option. Streambank stabilization, dredging, wetland restoration, and urban runoff management were ruled out for this study as all require surveys or field verification of sites for an accurate assessment. The WQT scenarios in this analysis center on conversion of corn and soybean row crop fields to perennial grassland, thereby reducing the input of fertilizers and export of nutrients to local waterways.

4.3. Water quality trading scenario development

4.3.1. Credit need

One of the first steps of the WQT process in Wisconsin is to determine the credit need of a point source. For each WWTF, this is the phosphorus[1] load reduction needed to meet their WQBEL. At the time of this work, no WWTF was meeting its WQBEL (Table 3) and many have either applied for or are operating under a variance. The WDNR sets phosphorus limits based on the type of receiving water that the WWTF discharges into. Eq. (1) shows the formula used to calculate annual total phosphorus (TP) credit need for each facility, where equation inputs are defined in Table 3.

$$\textbf{Credit Need}\left(\frac{\textbf{lbs}}{\textbf{yr}}\right) = \left(\textbf{TP}_{\textbf{avg}} - \textbf{TP}_{\textbf{WQBEL}}\right) \times \textbf{Q} \times 8.34 \times 365\left(\frac{\textbf{days}}{\textbf{yr}}\right) \quad (1)$$

Where:

TP_{avg} = Average effluent TP concentration (mg/L)

TP_{WQBEL} = Total phosphorus water quality-based effluent limit (mg/L)

Q = WWTF average flow rate in million gallons per day (MGD)

8.34 = Conversion factor ($L \cdot lb \cdot MG^{-1} \cdot mg^{-1}$)

4.3.2. Trade ratio

The WDNR requires a ratio to be applied to credits generated elsewhere in the watershed and used to satisfy the WWTF credit need (Eq. (1)). The ratio is required to account for uncertainty in modeling and actual management practice effectiveness on the ground. The minimum

[1] WQT is not limited to phosphorus and may be used to meet limits for various compounds. However, phosphorus limits set in facility permits are consistently not met, and numeric water quality criteria for phosphorus are readily provided for each facility, which is not the case for other pollutants such as nitrogen (WDNR 2020b).

trade ratio is 1.2 pounds of modeled phosphorus reduction in the watershed for each pound of phosphorus credit need at the WWTF. The ratio must be adjusted further for the following reasons:

1. Increased distance between trading partners
2. A credit generator is located downstream of credit purchaser
3. The traded pollutant is in different forms between trading partners
4. Uncertainty in nonpoint management practice effectiveness
5. Habitat restoration co-benefits

For all water quality trades in this study, no adjustment to the minimum ratio of 1.2 is necessary as (1) each field is within the same HUC-12 watershed as the partner WWTF, (2) each field is upstream of each partner WWTF, (3) the traded pollutant is in the same form between partners, (4) there is minimal uncertainty in the effectiveness of converting active cropland to perennial vegetation and (5) there is no habitat restoration co-benefit. When scenarios derived from the above conditions, such as when trading partners are located in different HUC-12 watersheds, WDNR provides guidance for how to include delivery factors to adjust the distance between the credit generator and the credit purchaser (WDNR, 2020c).

4.3.3. Modeled fields

The WDNR recommends trading within the same HUC-12 watershed to minimize the trade ratio when calculating phosphorus credits. ArcGIS was therefore used to identify the HUC-12 watershed (USGS 2022) associated with each WWTF, based on WWTF outfall locations provided by the WDNR.

Potential WQT fields (i.e., fields used to generate phosphorus reduction credits) in each WWTF HUC-12 watershed were identified as fields with crops that are common to the region or that have high fertilizer demands. Corn and soybeans are the most commonly planted field crops in Wisconsin, totaling 30.5 % and 22.0 % of the states cropland respectively (USDA 2017). Locations of crop fields were determined from reviewing the US Geological Survey (USGS) National Agricultural Statistics Service (NASS) Cropland Data Layer (CDL) (USDA NASS 2020) of each WWTF watershed. Using aerial imagery, CDL layers were reviewed for accuracy compared to actual field boundaries indicated by fence lines, roads, crop rows, and other boundary objects visible in aerial imagery.

Maps illustrating outfall location, HUC-12 boundary, and crop fields for the three archetype communities of Ontario, Viola and Wauzeka are available in Figures A1-S3, respectively.

4.3.4. SnapPlus model

SnapPlus was used to calculate the acres of corn or corn-soybean crop conversion to perennial grass required to satisfy the credit need of each WWTF. WDNR requires agricultural nonpoint source credit generators to use SnapPlus to calculate WQT credits. SnapPlus is a software program used throughout Wisconsin to prepare Natural Resources Conservation Service (NRCS) 590 Standard compliant nutrient management plans (NRCS WI 2015; WDNR 2018). Two critical features of SnapPlus related to water quality are its ability to generate, by field, a phosphorus loss value in pounds per acre per year, and capability to calculate soil erosion, based on the Revised Universal Soil Loss Equation, Version 2 (RUSLE2) (USDA-ARS, USDA-NRCS, the University of Tennessee, 1993).

Many model inputs, such as soil type, field slope, field length, and slope and length to perennial water are largely internally automated in the GIS mapping portion of SnapPlus called SnapMaps. However, default values are pulled from the soil survey database and sometimes deviate considerably from actual site conditions. Field slope is one such input that, during preliminary modeling, was found to have a large influence on erosion and corresponding nutrient export. Unfortunately, the only way to adjust field slope is to manually change the soil unit to one with an appropriate slope. However, there are other variables that are tied to soil unit that affect RUSLE2 calculations (albeit to a lesser extent than field slope), including slope length and erodibility factor. To address field slope discrepancies between SnapPlus default soil units and observed slopes (as calculated using actual digital elevation models in SWAT), we selected soil units that more closely matched actual field slopes while also trying to maintain consistency across other variables, to the extent practicable. Slope and length to perennial water variables were entered manually without having to choose a matching soil unit. Final SnapPlus inputs are provided in Table B1.

Important input variables that are not automated include soil characteristics, crop type, manure and fertilizer applications, and grazing activity. Typically, these inputs are determined through direct sampling or site visits. However, given the nature of this study, inputs were instead obtained from publicly available data sources, as described below.

The WQT fields used for this study are corn and soybean fields within each WWTF HUC-12 watershed, as identified using the National Agricultural Statistics Service Cropland Data Layer (NASS CDL). Repeated crop rotations were determined using three previous years of CropScape data (USDA NASS 2020). The field boundaries were digitized into shapefiles that were uploaded to SnapPlus. Table B2 provides the crop type for each field.

Conventional tillage practices and average crop yields were chosen for all fields. Tillage practices include fall chisel plowing, spring disking, and field cultivation before planting. It was assumed that manure application is not conducted on the WQT fields. This is a conservative assumption that results in an increase in the acreage needed to generate the credits needed to meet the WWTF's permit limit. Fertilizer applications were set to the recommended rates provided automatically in SnapPlus, which are based on Laboski et al. (2012).

Soil inputs were from statewide soil chemistry test results from the Wisconsin Department of Agriculture, Trade, and Consumer Protection (DATCP), collected from 1995 to 2014 (UW 2021). Data are available, by county, for phosphorus, potassium, organic matter, pH, calcium, magnesium, sulfur, zinc, and manganese. Average values for each field's county were used as inputs. If a field was located in multiple counties, the county that represented the majority of the field was chosen to represent the entire field. Soil inputs can be found in Table B3.

4.3.5. SWAT model

The SWAT model was used to model changes in nutrient exports that would result from conversion of row crops to grassland in the Kickapoo watershed. The Hydrologic and Water Quality System (HAWQS) Version 2.0 was used to set up and parameterize the Kickapoo SWAT model. Key spatial input files including DEM, land cover, soil map, and climate are provided in Table C1.

The watershed was partitioned into 92 sub basins at the Hydrologic Unit Code (HUC) 14 scale. These sub basins were further partitioned into 11,764 hydrologic response units (HRUs) with similar land use, soil, and slope classes.

In addition to the crop cover and agricultural management practices assumed for SnapPlus modeling, certain agricultural practice inputs were needed for SWAT modeling, including greater specification of tillage details and modeling of fields not associated with water quality trading (e.g., hay and pasture) for watershed calibration purposes. Inputs were informed by discussions with local stakeholders and extension agents as well as NRCS Standard 590 (NRCS WI 2015) and Wisconsin agronomy extension documents. Specific fertilizer, tillage, planting and grazing assumptions, by land cover, are provided in Table C2.

Daily discharge data from La Farge and Steuben USGS gage stations were used for calibration and validation of SWAT predicted streamflow. Daily data from the USGS Simplified Surface Energy Balance model parameterized for operational applications (SSEBop) (Senay et al. 2013) were aggregated to monthly values and used to validate predicted actual evapotranspiration at the Kickapoo watershed scale.

Weekly stream total phosphorus (TP) concentrations from Steuben

were used for TP load calibration from 2000–2020 using USGS Load estimator (LOADEST) (Runkel et al. 2004). Similarly, LOADEST was used to calculate monthly total nitrogen (TN) load from observed nitrate and total Kjeldahl nitrogen concentration data. La Farge was not included in LOADEST models due to insufficient nitrate and TP concentration data.

Full model setup and calibration results details are provided in Appendix C.

4.4. Life cycle inventory

This section describes the data sources and scope of developed life-cycle inventory (LCI) data. A complete LCI for each archetype system can be found in Appendix A, Section A2, documenting the inventory quantity, units, data source, data quality estimate, flow name and database for all flows included in the LCA model. LCIs for WWTF archetypes were used to calculate impacts and costs (Sections 2.5 and 2.6) on the basis of the study functional unit so that results could be extrapolated to non-archetype systems. A time horizon of 30 years was used for all system models.

4.4.1. WWTF inventory - baseline

Data from engineering design reports provided by the WDNR and pertaining to the Viola, Wauzeka and Ontario WWTFs were used to develop archetype models of aerated lagoon, activated sludge and RBC treatment systems, respectively, using CAPDETWorks wastewater treatment plant modeling software (Hydromantis 2014). CAPDETWorks is a software program intended for preliminary engineering design and cost estimates. Outputs from the CAPDETWorks software were standardized to the study functional unit and supplement facility specific data in each LCI. The software provides estimates of process electricity consumption, chemical use, sludge disposal requirements and major components of infrastructure such as earthwork, concrete and building space.

In some cases, values from the literature were used to estimate inventory items, notably for electricity demand associated with the aerated lagoon and activated sludge biological treatment processes. Estimates of nitrous oxide and methane greenhouse gas emissions released from biological treatment processes and surface water bodies receiving treated effluent were estimated based on the quantity of nitrogen and biochemical oxygen demand influent to each biological process and receiving waters (Intergovernmental Panel on Climate Change 2006). Inventory data sources are documented in Appendix A.

4.4.2. WWTF inventory – upgraded

Baseline WWTF inventories for most treatment processes remain consistent in the upgraded WWTF scenario. Engineering design calculations were used to estimate the required alum (chemical) dose necessary to reduce baseline phosphorus concentrations to meet Wisconsin's WQBELs using values specified in Table 3. Increased sludge production and processing requirements were included in each WWTF model by scaling inputs and outputs of affected unit processes linearly based on the increased quantity of sludge. A 26 % increase in sludge production was assumed. This value is based on survey results for activated sludge treatment plants adding chemical addition to achieve effluent total phosphorus concentrations of 1 mg/L or less (Stensel 2005).

Design and inventory data for the two-stage sand filter were based on standard design values assuming redundant construction and a filtration rate of 240 L/m^2-min (6 gpm/ft^2). This filter design includes two parallel filters, which can be alternately taken offline for backwashing while maintaining a recommended filtration rate of 120 L/m^2-min (3 gpm/ft^2). Depth of the sand media was assumed to be 0.76 m (2.5 feet) and is consistent with a conventional mono-media filter (Tchobanoglous et al. 2013). Inventory quantities for earthwork and basin concrete were estimated based on filter geometry with an assumed 40-year lifespan. The energy demand of sand filtration and rapid mixing of aluminum sulfate was assumed to be 0.2 and 0.3 kWh/m^3, respectively, based on the work of Arkhangelsky et al. (2017). The electricity demand of the sand filter assumes four meters of pressure head and includes the energy requirements for backwash pumping. No effect on effluent nitrogen concentrations is assumed in the analysis.

4.4.3. Agricultural field operations

For WWTF baseline and upgrade scenarios, in which corn and soybean row-crop practices would continue as usual, a simple LCI was developed for agricultural field operations. The purpose of this LCI was not to be comprehensive, but to capture typical and environmentally relevant impacts associated with typical agricultural practices.

Annual inputs of diesel fuel, nitrogen fertilizer and potassium fertilizer were estimated for row crops. We assume an average of 511/ha (5.5 gallons/acre) of diesel (Iowa State 2012; UNL 2024), 1057 kg/ha (113 lb/acre) of nitrogen fertilizer (N basis) and 430 kg/ha (46 lb/acre) of potassium fertilizer (K basis) (Laboski et al. 2012) per acre.

4.4.4. WQT inventory

The WQT scenario assumes that each WWTF will operate unchanged from the baseline scenario, that agricultural field operations and crop production will cease within a number of acres sufficient to generate the required phosphorus reduction credits, and that those acres will be converted to perennial grassland.

Following current regulatory requirements, SnapPlus was used to calculate the acres of fields that each WWTF community would need to convert (i.e., WQT fields) within their HUC12. For each community, average TP reductions per unit area were determined across all WQT fields then used to calculate a total required acreage based on the credit need and trade ratio (1.2). The WQT acres also influence the quantity of agricultural field operation inputs under baseline and upgrade scenarios.

SWAT model results were used to estimate the reduction in nutrient exports that result from converting corn or corn-soybean croplands to perennial grasslands across WQT fields (Table A9). LCI values were calculated by dividing these quantities by the annual average volume (m^3) of wastewater treated at the permitted facility.

To estimate the possible benefits of converting cropland in the Kickapoo watershed to a natural, unmanaged cover, we use data from Follett et al. (2012) who evaluated management effects on soil organic carbon (SOC) stocks across 14 prairie sites in the Great Plains region. Data relevant for this study are summarized in Table A10 and include average SOC concentrations and mass in the top 30 cm of soil. To translate data into an LCI format, we assume that conversion of cropland to perennial grassland will result in the accumulation of 15,186 kg SOC/ha of converted land in the top 30 cm of soil following recovery to 95 % of preagricultural levels. This value is based on a study of carbon accumulation after agricultural abandonment in the Minnesota sand plain, which predicted recovery would take 180–230 years (Knops and Tilman 2000). We assume 50 % recovery (7593 kg SOC/ha) would occur in the first 100 years, resulting in an annual accumulation of 76 kg SOC/ha/yr. Land areas reported in Table 1 were used in combination with this literature value on carbon accumulation to estimate the carbon sequestration benefit of the WQT option. More background and detail on carbon storage in agricultural systems and native prairie cover can be found in Section A2.3.2 of Appendix A.

4.4.5. Variability ranges

While one objective of this study is to model actual WQT scenarios, we also seek to explore the range of conditions under which WQT would or would not be advantageous from a cost or environmental impact perspective. Acknowledging there is uncertainty and variability in nearly every one of the LCI and cost line items, we focus here only on two sources of variability associated with agricultural nutrient emissions: predicted nutrient loadings from SnapPlus and SWAT. Although

some LCA results are influenced by both models' results, their variabilities are considered separately to avoid compounding of extreme values which would lead to unrealistic ranges.

Average areal TP reduction estimates from SnapPlus (Table 1) influence the amount of WQT acres necessary to meet credit requirements. WWTFs in watersheds where high reductions per acre can be realized will require fewer acres to be purchased and converted for credit. While less expensive, fewer environmental benefits would be accrued due to smaller decreases in agricultural field operations and smaller soil carbon sequestration potential. The opposite is true in watersheds where only low reductions can be realized. To evaluate the influence of the range of reductions predicted by SnapPlus (i.e., to imagine any one of the WWTFs is located in any one of the study watersheds), we used the range of average areal TP reductions (1.1–5.0 kg/ha/yr, Table 1) to establish a range of possible WQT acres necessary to meet credit requirements for each WWTF. This variability is illustrated by the error bars for all scenarios, costs, and impact categories except eutrophication potential of the WQT scenario, for which SWAT variability is considered, as discussed next.

Nutrient reductions estimated by SWAT are similarly variable to SnapPlus estimates and are a primary driver of eutrophication impacts under the WQT scenario. To illustrate the influence of this variability, we modeled the range of eutrophication impacts possible under the WQT scenario according to the range of average TP and TN reductions predicted by SWAT (1.2–2.8 kg TP/ha/yr, 3.4–5.7 kg TN/ha/yr, Table 2). This variability is illustrated by the error bars for eutrophication potential of the WQT scenario. WQT acre variability is not included here.

4.5. Life cycle impact assessment methods

Per ISO 14040, an LCA aims to be comprehensive in its selection of impact categories to identify potential trade-offs that may exist between impact categories. Results were generated as a part of this analysis are listed below. The LCI model was built and impact results were calculated using openLCA version 1.10.3 (GreenDelta 2020).

- Climate change potential (CCP) (Bare et al. 2002; Bare 2011).
- Cumulative energy demand (CED) (Althaus et al. 2010).
- Eutrophication potential (EP) (Bare et al. 2002; Bare 2011).

4.6. Life cycle costing methods

A LCCA was carried out for each archetype and each scenario (upgrade and WQT) designed to meet Wisconsin's WQBELs. No costs are included for the baseline scenario or operation of WWTF unit processes present in the baseline scenario as these costs are equivalent for all scenarios within a given archetype. Given this approach, the presented cost estimates do not reflect the cost of total system operation, but the additional cost associated with meeting the new water quality targets.

NPV was calculated over a 30-year period using Eq. (2) (equation 3–5 in Kneifel and Webb (2022)) assuming a base date of 2020. The LCCA was performed in constant (non-inflated) dollars and uses a real discount rate corresponding to the constant dollar method. The Department of Energy's 2022 recommended real discount rate of 3 % was used in this analysis (Kneifel and Lavappa 2022). In accordance with NIST guidelines, the LCCA only applies cost escalation factors to energy inputs. Electricity and natural gas costs were escalated according to energy escalation factors (excluding general inflation) for the commercial sector in the Midwestern U.S. (Kneifel 2022).

$$\boldsymbol{PV}\ (2020\ \$) = \frac{\boldsymbol{F_t}}{(1+\boldsymbol{d})^t} \tag{2}$$

Where:
PV = Present value (2020 $)
F_T = Future value at time t
t = time (year)
d = discount rate

Engineering feasibility reports were the primary source of information for the LCCA of technological upgrades at WWTFs. Cost estimates in the feasibility reports are representative of the period between 2016 and 2018. Input cost data were adjusted to the base year using the RSMeans Construction Cost Index. Cost estimates for the WQT scenarios were based on previously completed WQT projects in Wisconsin by the company of one of this study's co-authors, so are reflective of current market conditions.

The LCCA includes capital and annual costs. Annual costs are categorized as operation costs, material costs, chemical costs, and energy costs. No known sources of revenue are associated with any of the project scenarios. Interest on capital expenditures is excluded from the analysis based on the assumption that communities can receive funding through WDNR's Clean Water Fund Program which offers subsidized, 0 % interest rates, for extremely disadvantaged municipalities with populations of less than 1,000 or with less than 65 % of Wisconsin's median household income (WDNR 2022a).

4.6.1. Cost of WWTF upgrade

Capital costs associated with adding sand filtration and chemical addition to the lagoon-based treatment system include construction of effluent filters, chemical mixing equipment, an additional final clarifier, sludge storage structure, and associated buildings. The activated sludge system already has a final clarifier where flocculation and settling can occur following chemical addition, as well as sludge storage facilities. Capital costs for the activated sludge system therefore only included construction of the effluent filters and chemical mixing equipment.

Annual costs include general operation and maintenance (O&M) costs, chemical cost, electricity, and a fee for additional sludge disposal.

Cost estimate data specific to the RBC treatment system were not available. Cost data for the activated sludge plant was used as proxy, scaling cost data linearly based on the ratio of design and average flow rates of the RBC and activated sludge systems for capital costs and annual costs, respectively. This proxy is expected to be reasonable for this level of assessment given that the same infrastructure additions are required, the size of both facilities is similar, and both facilities already have an existing final clarifier and similar sludge handling practices (aerobic digestion).

Table A12 (supplemental information) lists the cost data associated with each treatment alternative.

4.6.2. Cost of WQT

WQT costs are borne by the WWTF, making them comparable to technological upgrade costs. Current estimated WQT costs are provided in Table A11. They are based on the experience of current market participants who have developed WQT plans and facilitated trading in other watershed. Costs include:

- Development of a WQT Plan, permitting through WDNR, and negotiation of credit costs with landowners;
- One-time farming activities to convert fields to perennial vegetation;
- Annual costs to purchase credits and inspect and maintain vegetation after planting.

Table A12 (supplemental information) lists the cost data values associated with WQT for each case-study community.

Disclaimer

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

Data will be made available on request

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